

# INT221:MVC PROGRAMMING

L:2 T:0 P:2 Credits:3

**Course Outcomes:** Through this course students should be able to

CO1 :: recall features of Laravel and the installation process.

CO2 :: apply routing, handle HTTP requests, and customize responses for seamless Laravel application navigation.

CO3 :: develop controllers, utilize Blade templates, and apply advanced routing for modular and structured development.

CO4 :: manage cookies, emails, sessions for convenient and robust functionality

CO5 :: design forms and execute form validation to ensure data integrity

CO6 :: employ databases to streamline data handling and enhance functionality in Laravel.

## Unit I

**Getting started with MVC Laravel framework** : What is MVC framework?, Overview of Laravel framework and its features, Introduction to composer, Latest Composer installation, Latest Laravel installation, Directory/Application structure, Artisan

## Unit II

**Request, Routing & Responses** : Laravel Request Lifecycle, Basic Routing, Routing Parameters, Understanding Views in Laravel, Passing Data to Views, Sharing Data with all Views, Laravel Response- Attaching Headers, Attaching Cookies, JSON Response, Laravel Redirections, Redirecting to Named Routes, Redirecting to Controller Actions

## Unit III

**Controllers, Blade and Advanced Routing** : Basic Controllers- Creating Controllers, Controller Routing, Controller Middleware, Restful Resource Controllers, Blade- Creating Templates, PHP Output, Controller Structures, Templates Inheritance, Advanced Routing- Named Routes, Secure Routes, Parameter Constraints, Route Groups, Route Prefixing, Domain Routing, URL Generation- The current URL, Generating Framework URLs, Asset URLs, Generation Shortcuts

## Unit IV

**URL Generation, Request Data and Emails** : Request Data-Retrieval, Old Input, Uploaded Files, Cookies, Sending Emails

**Laravel Localization and Sessions** : Laravel Localization and Examples, Laravel Session- Accessing Session Data, Storing Session Data, Deleting Session Data

## Unit V

**Laravel Form validation** : CSRF field, Method field, Laravel Form validation, Validation Rules, Error Messages, Custom Validation Rules, Repopulating Forms

## Unit VI

**Getting started with databases** : Model creation, Migrations, CRUD using Query Builder, Seeding, Using MongoDB with Laravel, CRUD using Eloquent ORM(Object Relational Mapper), Implementing Rest APIs

## List of Practicals / Experiments:

### List of practicals

- Installation of Laravel and Composer using command prompt
- Implement sharing of data using views
- Use route parameters and parameter constraints
- Implement redirections using routes and controllers
- Create and register middlewares in Laravel
- Create blade templates, controller structures and template inheritance in Laravel
- Use restful resource controllers to perform CRUD operations
- Working with advanced routing and URL methods in Laravel

- Send emails and upload files in Laravel
- Access, store and delete cookie and session data in Laravel.
- Create form and implement simple validation, error messages, custom validation rules and custom validation messages in Laravel.
- Connect to database in Laravel
- Create, modify, update tables in Laravel
- Manage database with migrations

**References:**

1. ULTIMATE LARAVEL FOR MODERN WEB DEVELOPMENT by DRISHTI JAIN, AVA PUBLISHING